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Effect of Internal Magnetic-Field Gradients on Nuclear-Magnetic-Resonance Measurements and Nuclear-Magnetic-Resonance-Based Pore-Network Characterization

Saurabh Tandon and Zoya Heidari, University of Texas at Austin

Summary

Nuclear-magnetic-resonance (NMR) measurements have been extensively used for determining porosity, pore-size distribution, and fluid saturation in porous media. However, internal gradients of the magnetic field generated by the presence of paramagnetic or diamagnetic centers such as shale or clay particles can significantly affect NMR response. Consequently, the resulting interpretation for pore-size distribution and porosity is also affected. In this paper, we quantify the effect of internal magnetic-field gradients and spatial distribution of matrix components such as clay minerals on NMR response using pore-scale NMR numerical simulations. We also quantify the influence of the aforementioned parameters on the NMR-based evaluation of porous media.

We used the finite-volume method to numerically solve the Bloch (1946) equations and simulated magnetization decay in porous media. We cross validated the reliability of numerical simulations using analytical solutions given for spherical pores in different diffusion regimes. The model was then used for simulation of NMR response in the pore-scale images of sandstone and carbonate rocks. We used Larmor frequency of 2 MHz, external magnetic-field gradient of 0.10 T/m, and half-echo spacing time of 0.5 ms for simulating NMR response in pore-scale images of sandstones and carbonates. We then developed synthetic cases using actual rock images covering a wide range of spatial distribution of clay minerals (i.e., paramagnetic centers) to quantify the sensitivity of NMR decay to internal magnetic-field gradients. We quantified the sensitivity of NMR response for distribution of clays as thin laminae in the rock and as thin layers on the surface of the grains.

The results showed that at low concentration (0.3 to 0.7%) of dispersed clay, there is a negligible effect of internal magnetic-field gradients on magnetization decay. At higher concentration of dispersed clay (5.1 to 7.3%), we observed a significant effect of internal magnetic-field gradients on magnetization decay. The presence of shale minerals can cause 53% variation in the location of the transverse-relaxation-time constant (T_2) and up to 67% relative error in the assessment of dominant pore sizes. Shale laminations containing clay were found to produce an effect of up to 31.8% on T_2 relaxation-time constant, which could cause a relative error of 50.0% in estimates of dominant pore size in the rock.

The outcomes of this paper demonstrated the effect of heterogeneous rock mineralogy on NMR response. The effect of internal magnetic-field gradients generated by shale and clay on NMR becomes relevant when shale and clay particles are close to the pore fluid and their magnetic field starts to affect the distribution of magnetic field in the pore space. The results reveal the importance of characterizing the distribution of shale and clay minerals before interpreting NMR response and can potentially improve conventional techniques of pore-network characterization (pore-size distribution and pore volume) in the presence of clay minerals where internal magnetic-field gradients are not negligible.

Introduction

NMR studies of porous media have been used to obtain lithology-independent estimates of porosity, pore-size distribution, rock permeability, and fluid saturations (Toumelin et al. 2002, 2007; Mohnke and Klitzsch 2010; AlGhamdi and Arns 2012). NMR measurements comprise applying magnetic fields and constant magnetic-field gradients to quantify dephasing of proton spins in pore space (Torrey 1956; Bergman and Dunn 1995; Hürlimann and Griffin 2000). However, NMR response becomes complicated because of the presence of susceptibility differences in pore fluid and matrix components that produce internal magnetic-field gradients (Brown 1961; Dunn 2002; Mohnke and Klitzsch 2010; AlGhamdi and Arns 2012). Internal magnetic-field gradients distort the externally applied magnetic fields (Dunn 2002; Sun and Dunn 2002). The effect of internal magnetic-field gradients on externally applied magnetic fields can be significant if paramagnetic or diamagnetic clays with high susceptibilities are present in the matrix (AlGhamdi and Arns 2012; Burcaw and Callaghan 2012). Pore-scale simulation of NMR could help in quantifying the effects of internal magnetic-field gradients, generated by magnetic-susceptibility contrast, on NMR measurements and could help in improving NMR interpretation when this effect is significant.

The movement and dephasing of spins in a medium in a laboratory frame of reference during NMR relaxation is governed by the Bloch (1946) and Torrey (1956) equations, given by

$$\frac{d\mathbf{M}}{dt} = \gamma(\mathbf{M} \times \mathbf{B}) - \frac{\mathbf{M} - \mathbf{M}_0}{T_B} + D\nabla^2\mathbf{M}, \quad \dots \dots \dots (1)$$

where $\mathbf{M}$ is magnetization, t is time, $\mathbf{B}$ is the total magnetic-field vector, γ is the gyromagnetic ratio, $\mathbf{M}_0$ is the initial magnetization, D is the diffusion tensor, and T_B is the bulk-relaxation-rate tensor (Torrey 1956). The precession frequency is governed by the Larmor precession equation, given by

$$f_0 = \gamma B_0, \quad \dots \dots \dots (2)$$

where f_0 is the precession frequency and B_0 is the external static magnetic field. For simplicity, the laboratory frame of reference can be shifted to a rotating frame of reference (Toumelin et al. 2007; Mohnke and Klitzsch 2010). This reference frame rotates at Larmor frequency. The diffusion tensor can also be expressed as a single diffusion constant, if the diffusivity is the same in all directions. The bulk-relaxation-rate tensor comprises transverse bulk-relaxation rate (T_{2B}) in the transverse plane (i.e., x - y plane) and longitudinal bulk-relaxation rate (T_{1B}) in the z -direction. Eq. 1 can be decomposed into three separate equations for each component of magnetization, given by

$$\frac{dM_x}{dt} = \gamma(M_y B_{\text{eff}}) - \frac{M_x}{T_{2B}} + D \nabla^2 M_x, \quad \dots \dots \dots (3)$$

$$\frac{dM_y}{dt} = \gamma(M_x B_{\text{eff}}) - \frac{M_y}{T_{2B}} + D \nabla^2 M_y, \quad \dots \dots \dots (4)$$

$$\frac{dM_z}{dt} = \frac{M_0}{T_{1B}} - \frac{M_z}{T_{1B}} + D \nabla^2 M_z, \quad \dots \dots \dots (5)$$

where B_{eff} is the effective magnetic field that is the superposition of externally applied and internally induced static magnetic fields in the rotating reference frame; D is the diffusion constant; and M_x , M_y , and M_z are the x -, y -, and z -component of magnetization, respectively. In formulating Eqs. 3, 4, and 5, we assume that magnetic-field gradients in the x - and y -direction are negligible compared with magnetic-field gradients in z -direction. It is also assumed that the initial magnetization is applied only in the z -direction. In the case of low-field NMR spectrometry of protons, the Larmor frequency is assumed to be approximately 1 to 2 MHz. This gives the external static magnetic field, B_0 , of 250–470 gauss. Such low frequencies are often used both in logging tools and the laboratory-scale studies for measurement of magnetization decay (Prammer et al. 1998; Freedman et al. 2003; Johnston et al. 2003; AlGhamdi and Arns 2012; Alhwety et al. 2014; Boitel et al. 2015). The initial magnetization generated in the sample is also assumed to be parallel to the z -direction and is calculated using

$$M_0 = x_w B_0, \quad \dots \dots \dots (6)$$

where x_w is the magnetic susceptibility of the fluid and B_0 is the external static magnetic field, applied only in the z -direction.

Eqs. 3, 4, and 5 can be solved analytically or numerically with suitable boundary conditions to provide evolution of magnetization in a system. Torrey (1956) solved the system analytically to give the formulation of free induction decay for protons. Carr and Purcell (1954) solved the system analytically, using the central-limit theorem to obtain a Gaussian distribution of proton spins. They considered continuous pulses being applied with a constant time interval between two refocusing pulses, defined as echo time by Carr and Purcell (1954). They developed the Carr-Purcell MeiBoom-Gill (CPMG) sequence that attenuates the effect of the local magnetic fields on the fluid protons. The CPMG sequence consists of applying a constant magnetic field to develop initial magnetization. Then a 90° focusing pulse (90_y) in the y -direction is applied to z magnetization to rotate it in the transverse plane. After applying an initial 90° pulse, a series of 180° refocusing pulses (180_x), orthogonal to the initial 90° pulse, are applied for the entire duration of the experiment. Continuous application of refocusing pulses, with a constant time interval between two refocusing pulses, will refocus the transverse magnetization along the x -axis periodically, and the influence of local magnetic fields on magnetization decay can be ignored (Mohnke and Klitzsch 2010).

Brownstein and Tarr (1977) proposed the mixed boundary condition that governs the decay of spins on the boundary surfaces of porous media and introduced the theoretical basis for surface relaxation. The exact formulation of the boundary condition is given by

$$D \nabla \mathbf{M} \cdot \frac{\mathbf{n}}{l} - \frac{\mathbf{M} - \mathbf{M}_0}{T_{1s}} \Big|_{\Omega} = 0, \quad \dots \dots \dots (7)$$

where $\mathbf{n}$ is the unit vector normal to the boundary face Ω ; l is the length scale where the boundary condition operates; and T_{1s} is the relaxation rate for the boundary (Cohen and Mendelson 1982). The surface relaxivity is defined as

$$\rho = \frac{l}{T_{1s}}, \quad \dots \dots \dots (8)$$

where ρ is the surface relaxivity at the boundary.

Brownstein and Tarr (1979) provided solutions for transverse magnetization decay in simple geometries and introduced the idea of inversion of magnetization decay to understand the behavior of eigenvalues for the formulation. Three relaxation regimes, including surface-limited regime, intermediate regime, and diffusion-limited regime, were defined on the basis of the behavior of eigenvalues. They showed that in a porous medium the inversion of magnetization decay could be correlated to the surface/volume ratio of the porous media, and the decay constant for the surface-limited regime can be formulated as

$$\frac{1}{T_{1/2}} = \frac{1}{T_{1B/2B}} + \rho \frac{S}{V}, \quad \dots \dots \dots (9)$$

where $T_{1/2}$ is the longitudinal or transverse relaxation rates; S is the surface area of the medium; and V is the volume of the medium.

Internal magnetic-field gradients are generated by magnetic-susceptibility contrasts between the pore fluid and the porous media (Hürlimann 1998; Ryu 2009). The effects of internal magnetic-field gradients depend on the shape and size of the pores in addition to the applied external magnetic field and susceptibility contrast between the pore fluid and rock matrix (Burcaw and Callaghan 2012). In the presence of internal and external magnetic-field gradients, a dephasing term has to be introduced to account for these gradient effects. Torrey (1956) analytically solved Eq. 1 for proton-spin dephasing caused by magnetic-field gradients in an infinite medium and showed

$$\frac{1}{T_D} = \frac{2}{3} D (\gamma G \tau)^2, \quad \dots \dots \dots (10)$$

where T_D is the diffusion-induced time-decay constant; G is the magnetic-field gradient; and τ is one-half of the echo time between two consecutive 180° refocusing pulses in the CPMG sequence. The formulation of Eq. 10 is valid when the time required for the proton spins to move to the pore walls is much larger than the time required for the spin to dephase in the externally applied magnetic field (Torrey 1956; Hürlimann 1998). The exact solution for explaining the variation of the diffusion-decay constant for different pore shapes and echo spacing was discussed by Neuman (1974).

Brown (1961) introduced the idea of generation of internal magnetic-field gradients by randomly placed dipoles in an effective medium. Bergman and Dunn (1995) used periodic sphere packing with existing dipoles and discussed the effect of diffusion in the presence of internal magnetic-field gradients. Audoly et al. (2003) and Burcaw and Callaghan (2012) assumed a distribution of spherical dipoles for understanding heterogeneity in porous media. The magnetic field generated at a point a distance r away from a sphere of radius R_0 and magnetic permittivity μ_g , located in a medium of magnetic permittivity μ_w , is given by

$$B(r) = \frac{\mu_g - \mu_w}{\mu_g + 2\mu_w} B_0 R_0^3 \left(\frac{3z^2}{r^5} - \frac{1}{r^3} \right), \dots \dots \dots (11)$$

where z is the distance parallel to the z -coordinate of the random point away from the sphere center (Brown 1961; Audoly et al. 2003). The overall magnetic-field strength is given as the superposition of fields imposed by individual dipoles.

Hürlimann (1998) reviewed three different diffusion regimes (i.e., short time, motionally averaging, and localization) using the magnetization decay in a constant magnetic-field gradient field. He presented the criteria for distinguishing different diffusion regimes as well as the analytical solutions for the magnetization decay for each of these regimes. Anand and Hirasaki (2007) studied the internal magnetic-field gradients generated by a coating of paramagnetic particles on the surfaces of spheres using random-walk simulations. AlGhamdi and Arns (2012) used the random-walk method to simulate NMR magnetization decay in pore-scale computed-tomography (CT) images of both sandstone and carbonate-rock samples and included internal magnetic-field gradients generated by susceptibility contrasts caused by both matrix components and paramagnetic particles present in the rock.

There is no systematic study on the effect of shale distribution in rocks on NMR response using micro-CT images of rock samples documented in previous publications. Shale could be present in the rock matrix in the forms of structural shale, as dispersed (grain-coating) shale on the surfaces of grains, or as shale laminations (Thomas and Stieber 1975; Moss 2008). The effect of shale minerals on NMR measurements depends on the type of shale, its distribution, and pore-size distribution of the rock.

Hirasaki et al. (2001) studied the effect of clay on internal magnetic-field gradients on North Burbank and Berea sandstones using experiments as well as simulations. They used a spherical pore model and a simple model for generation of magnetic-field gradients to capture the effect of grain-coating clay particles covering the surface of the pores. They did not make any distinction between shale and clay in their study. They concluded internal magnetic-field gradients derived from chlorite in North Burbank sandstone cause a more-significant error in NMR T_2 response than in Berea sandstone. Moss (2008) experimentally investigated the effect of increasing shale on NMR measurements in different rock samples and documented the variation in NMR response caused by different configurations of shale. However, the author did not explicitly discuss the effect of shale internal magnetic-field gradients on T_2 distribution.

AlGhamdi and Arns (2012) and Arns et al. (2011) modeled the NMR response in the presence of shale on micro-CT images of sandstones and carbonates. They also did not distinguish between shale and clay explicitly in their publications and did not investigate the sensitivity of the NMR response to changes in shale concentration. Alhwety et al. (2014) investigated the effects of laminations, contrast in magnetic susceptibility of grains, and diffusional coupling for laminated systems using synthetic rock samples created with a Boolean process. The work uses simulated rock samples with different pore-size distributions and then investigated the change in NMR response with increasing values of magnetic susceptibility. Anand (2007) experimentally investigated the sensitivity of NMR to magnetic particles dispersed on the grain surfaces. In that work, silica grains are used as model sandstones. The silica-grain surfaces were coated with magnetic particles by flowing solutions with different concentrations of magnetic nanoparticles, and changes in NMR T_2 response were studied.

In this paper, we systematically study the effect of internal magnetic-field gradients generated by matrix components including grains and shale on NMR T_2 distribution in sandstone and carbonate rocks. We numerically simulate NMR magnetization decay and calculate T_2 distribution in pore-scale images from sandstone and carbonate-rock samples. We then create synthetic examples using real images with different spatial distributions of shale. The results of numerical simulations quantify the influence of the shale distribution on NMR interpretation results.

Method

This section describes the numerical-simulation method and the development of synthetic samples used for quantifying the effects of shale distributions and internal magnetic-field gradients on NMR measurements. We first describe numerical simulation of the Bloch (1946) and Torrey (1956) equations (Eq. 1) and verification of the developed simulator. Then, we state the processing of micro-CT images of sandstone and carbonate-rock samples and the development of synthetic cases with different types of shale distributions used for sensitivity analysis. We finally simulate the NMR response in micro-CT images and synthetic samples to investigate the effects of shale topology on NMR T_2 distribution.

Numerical Modeling of Bloch (1946) and Torrey (1956) Equations. We modified an existing finite-volume simulator OpenFOAM (Weller et al. 1998) to numerically solve Eq. 1 with the boundary condition given in Eq. 7. The simulation code was verified by comparing the estimated magnetization decay with the analytical solution given for a spherical pore provided by Brownstein and Tarr (1979) and Kleinberg and Horsfield (1990).

Sample Selection and Processing of Micro-CT Images. We simulated the NMR response for two sandstone (i.e., S1 and S2) and two carbonate (i.e., C1 and C2) samples. In the case of Sample S1, Fiji, an open-source image-processing software (Schindelin et al. 2012) was used to convert the original gray-scale micro-CT images to a binary image. Then, ImageJ, an open-source image-processing program (Schindelin et al. 2012) developed at the National Institutes of Health, was used to reconstruct 3D rock matrices. The micro-CT images of sandstone Sample S2 and carbonate Samples C1 and C2 were scanned and digitized by the Petroleum Engineering and Rock Mechanics Group at Imperial College London (Dong 2007). **Table 1** lists properties of rock samples used in the simulations. **Figs. 1a through 1d** show the binary 3D micro-CT scan images of the four rock samples.

Rock Sample	Resolution ($\mu\text{m}/\text{pixel}$)	Image Porosity	Sample Size (pixel^3)	Surface Area (μm^2)
S1	1.80	0.154	100×100×100	1.4×10^5
S2	2.42	0.127	100×100×100	2.0×10^5
C1	2.68	0.179	100×100×100	2.6×10^5
C2	2.65	0.201	100×100×100	2.1×10^5

Table 1—Image properties of the rock samples used as inputs to numerical simulations.

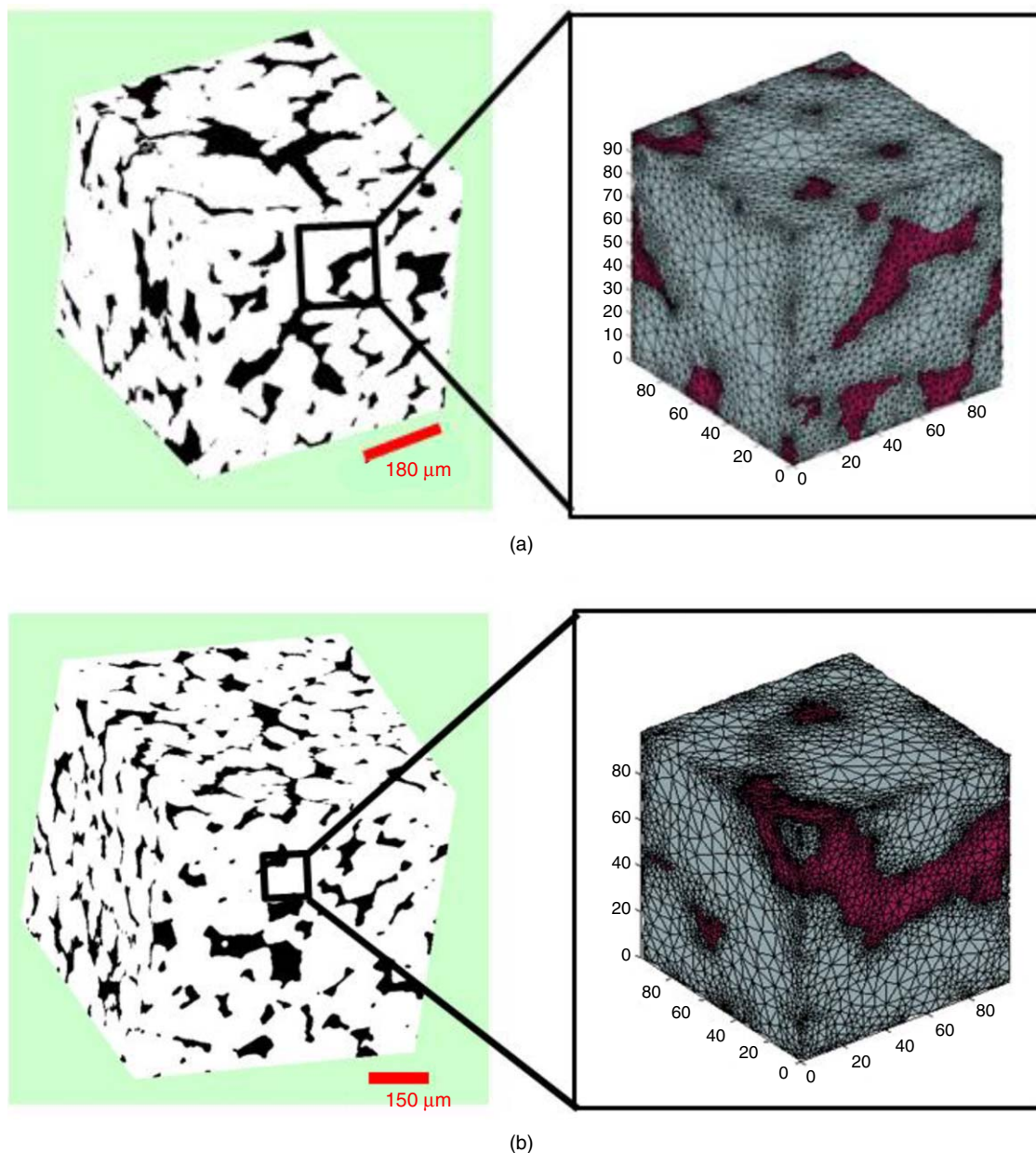


Fig. 1—The 3D pore-scale images of the four rock samples used as inputs to numerical simulations: (a) sandstone Sample S1, (b) sandstone Sample S2, (c) carbonate Sample C1, and (d) carbonate Sample C2.

We used iso2mesh and gmsh (Fang and Boas 2009; Geuzaine and Remacle 2009) open-source Matlab and C++ libraries for generating tetrahedral meshes in the gray-scale images. After generating the mesh, we created several synthetic cases using the processed real images covering a wide range of paramagnetic- and diamagnetic-susceptibility contrasts among the grains, shale, and pore fluids to investigate their effects on NMR measurements.

We first introduced the susceptibility contrast between the rock grains and the pore fluids. This susceptibility contrast introduces internal magnetic-field gradients in the pore space. The magnetic-dipole centers were assumed to be the centroids of each element of the grain mesh. The volume of each dipole was assumed to be the volume of each tetrahedral element of the mesh. The internal magnetic field was calculated using Eq. 11 for each point in the pore space. The calculated internal magnetic field is then used as input for numerical simulation of magnetization decay.

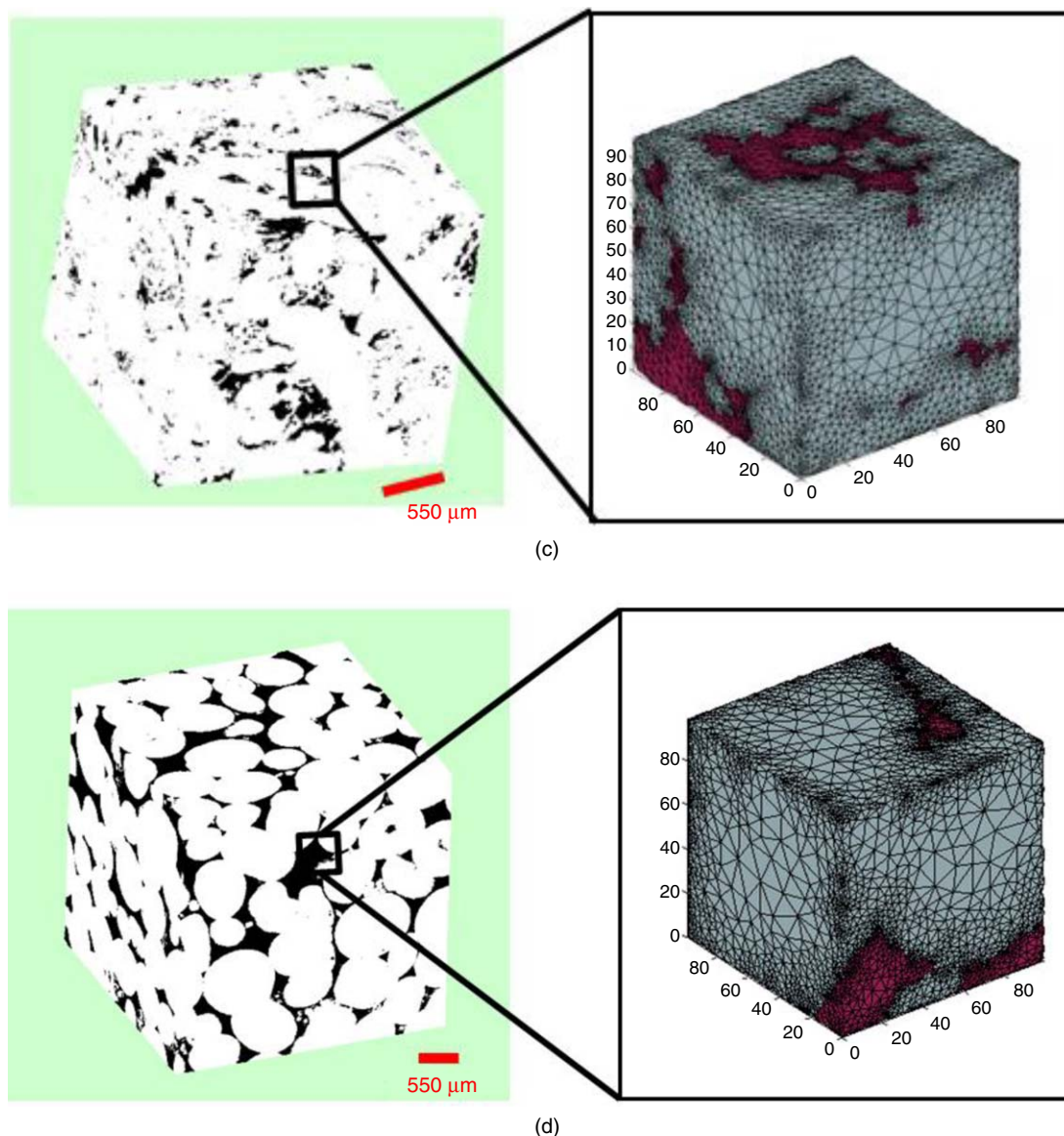


Fig. 1 (continued)—The 3D pore-scale images of the four rock samples used as inputs to numerical simulations: (a) sandstone Sample S1, (b) sandstone Sample S2, (c) carbonate Sample C1, and (d) carbonate Sample C2.

Synthetic samples with ideal distributions of shale in the pore space were generated from the available sandstone and carbonate pore-scale images to investigate the effect of shale on NMR measurements. This paper includes the simulation results in two ideal cases: (1) dispersed shale consisting of 60% volumetric concentration of clay and (2) shale laminations consisting of 60% volumetric concentration of clay. To identify the effect of only internal magnetic-field gradients generated by clay minerals in shale on magnetization decay, the susceptibility contrast between the diamagnetic grains, nonclay components of shale, and pore fluids was assumed to be negligible, and the presence of shale was assumed not to have any effect on the surface relaxivity of the rock grains.

Dispersed Shale. The first synthetic case comprises a layer of dispersed (grain-coating) shale distributed on the surface of the grains. The dispersed shale contains both paramagnetic clay and nonclay components. The dispersed shale is assumed to contain 60 vol% concentration of paramagnetic clay. The nonclay components of shale are assumed to have negligible magnetic-susceptibility contrast with pore water and hence do not create any internal magnetic-field gradients. The changes in total rock porosity caused by interstitial porosity of the dispersed shale have been assumed to be minimal (0.1 to 1.4%, 0.1 to 0.8%, 0.2 to 0.9%, and 0.1 to 0.7% for Samples S1, S2, C1, and C2, respectively). The shale layer is also assumed to not significantly affect the pore volume and the topology of the pore space to isolate the effects of internal magnetic-field gradients on NMR measurements.

The thickness of the shale layer was assumed to increase from 0 to 5000 nm. To isolate the sensitivity of NMR T_2 measurements to the thickness of the shale layer, we kept the composition of the shale layer constant for different shale-layer thicknesses. The volume of each shale dipole was assumed to be proportional to the volume of shale present on the grain surface. Superposition of magnetic fields, exerted by individual dipoles, provided the total magnetic field in the pore space. The z -gradient of the total magnetic field was used as input for numerical simulation and was used for quantifying the effect of the internal magnetic-field gradients on NMR measurements.

The overall solid volumetric concentration of clay minerals is defined as the ratio of the volume of clay minerals to the total volume of shale and nonshale matrix components in the entire sample (comprising both the shale and the grains). **Table 2** lists the solid volumetric concentration of clay in the synthetic cases, developed using Samples S1, S2, C1, and C2.

Sample No.	Overall Volumetric Concentration of Clay (%)		
	Shale-Layer Thickness: 500 nm	Shale-Layer Thickness: 2500 nm	Shale-Layer Thickness: 5000 nm
S1	0.7%	3.6%	7.3%
S2	0.5%	2.6%	5.1%
C1	0.5%	3.0%	5.5%
C2	0.4%	2.0%	4.1%

Table 2—Solid volumetric concentration of clay minerals for different cases of clay-layer thickness (i.e., 500, 2500, and 5000 nm) in rock Samples S1, S2, C1, and C2.

Laminated Shale. The second synthetic case comprises a shale lamina made of clay, small pores containing bound water, and nonclay components in the rock. The volume of shale lamina increases from 5 to 20% of the volume of each rock sample. **Fig. 2** shows the lamina generated in Samples S1, S2, C1 and C2. Clay particles were assumed to be randomly distributed in the shale lamina. Shale laminations consisted of 60% volumetric concentration of clay. The clays in shale laminations are assumed to be randomly distributed.

The pores in lamina were also assumed to be randomly distributed. Each pore is an effective combination of many micro-sized pores present in the shale sample. The surface relaxivity of the pores and the bulk-relaxation-rate constant were assumed to be 10 $\mu\text{m/s}$ and 0.1 seconds, respectively (AlGhamdi and Arns 2012). We assumed that the water in shale pores is in a short-time diffusion regime and does not undergo any dephasing caused by internal magnetic fields (Hürlimann 1998). Laminations also introduced changes in the pore structure as well as in the porosity of the samples. **Table 3** lists the change in sample porosity with increase in the volume of shale lamination.

Simulation of Magnetization Decay. The pore-scale images obtained from micro-CT and the synthetically generated samples were inputs to the numerical simulator. **Table 4** summarizes the assumed parameters used as inputs for numerical simulations of the Bloch equations. The values assumed for the frequency of static magnetic field (470 gauss), half-echo spacing time (0.5 ms), and external magnetic-field gradient (0.1 T/m) are within the typical range of values that are used for NMR logging and laboratory work (Prammer et al. 1998; Freedman et al. 2003; Johnston et al. 2003; Toumelin et al. 2007; AlGhamdi and Arns 2012; Jarath, Torres-Verdin, and Merletti, G. 2012; Alhwety et al. 2014; Boitel et al. 2015; Wolanski et al. 2015). The use of such high values for half-echo time (i.e., 0.5 ms) and externally applied magnetic-field gradient provides an insight into the effect of internal magnetic-field gradient on different experiments and pulse sequences that use high static magnetic fields or different magnetic-gradient strengths and echo times. Examples of such cases include pulse-field-gradient sequences (Tanner 1965) and diffusion-editing sequence (Flaum 2007).

For simulations, it was assumed that the initial wait time was large enough for complete initial buildup of magnetization. Then, a 90_y pulse was applied that rotated magnetization into the transverse plane. The initial 90_y pulse was followed by a series of 180_x pulses to generate the CPMG sequence. Each 180_x pulse was assumed to completely rotate the magnetization with respect to the x -axis. For a very small timestep, large pore radii, and small echo-spacing time, the rock pores could be assumed to be in free-diffusion regime (Hürlimann 1998). The formulation provided by Torrey (1956) and Carr and Purcell (1954) remains valid, and magnetization decay caused by internal or external magnetic-field gradients could be simulated using Eq. 10. The overall magnetic-field gradients in the rock were modeled through superposition of externally applied magnetic-field gradients and internal magnetic-field gradients generated by susceptibility contrasts between rock matrix components and pore fluids.

Inversion of Magnetization Decay. T_2 distribution was obtained by inversion of magnetization decay using the Levenberg-Marquardt fitting method and the zeroth-order curve-fitting scheme (Dunn et al. 2002; Aster et al. 2005). The inversion method used for obtaining T_2 distribution from magnetization decay is described by Medelín et al. (2016).

Results

Code Validation. The simulations were first validated by simulating magnetization decay for a spherical pore with bulk- and surface-relaxation effects and no magnetic-field gradients. **Fig. 3** shows the comparison of simulated magnetization decays for spherical pores of different pore radii against the analytical solution provided in Brownstein and Tarr (1979) for this simplified case. The radius of the spherical pore, R_s , was increased from 10 to 150 μm to cover different relaxation regimes defined by Brownstein and Tarr (1979). Appendix A describes the analytical solution for this simplified case. In the case of a single spherical pore with surface relaxivity of 5 $\mu\text{m/s}$ and transverse bulk-relaxation rate of 2.5 seconds, the difference between the results from the numerical simulator and the analytical method in all cases was less than 5%. Each simulation took a few minutes to run on a desktop computer with a single core and 8 GB RAM.

We then validated the simulation results by simulating magnetization decay for a spherical pore under a constant magnetic-field gradient with CPMG echo sequence and no surface effects. The radius of the spherical pore, R_s , was increased from 10 to 150 μm . Neuman (1974) and Kleinberg and Horsfield (1990) discussed the analytical solution for this simple case. The analytical solution is presented in Appendix B. **Fig. 4** shows the comparison of simulated magnetization decays with the analytical solution. For each of these cases, the externally applied magnetic-field gradient is assumed to be 0.1 T/m and the half-echo time (τ) is assumed to be 0.0005 seconds. The 90_y pulse and 180_x pulses in the CPMG echo sequence are assumed to perfectly rotate the magnetization vector. The pores are assumed to have reflecting walls and a fluid bulk decay constant of 2.5 seconds. The error in each case is found to be less than 5%.

We finally validated the simulation results by simulating magnetization decay for a spherical pore under a constant magnetic-field gradient with CPMG echo sequence and surface effects modeled using Eq. 7. The half-echo times assumed for these simulations are 0.001 seconds. All the other rock properties and simulation parameters are assumed to be the same as those from the previous verification case (Figs. 3 and 4). **Fig. 5** shows the comparison of simulated magnetization decays with the analytical solution. We observed excellent agreement between the simulated and the analytical solution in all the cases.

Effect of External Magnetic-Field Gradients on NMR T_2 Distribution. We simulated the NMR T_2 distribution for Samples S1, S2, C1, and C2 in two cases under different assumptions. In the first case, we ignored magnetization decay caused by the externally applied magnetic-field gradients. In the second case, we accounted for magnetization decay caused by the externally applied magnetic-field

gradients. The internal magnetic-field gradients generated by susceptibility contrasts are assumed to be negligible in both cases. Table 4 lists the assumed echo time and the externally applied magnetic-field gradient. Fig. 6 shows the simulated T_2 distribution demonstrating the effect of externally applied magnetic-field gradients on Samples S1, S2, C1, and C2. We observed that the T_2 peak moves to the left when the magnetization decay caused by externally applied magnetic-field gradients is taken into account. The simulation results showed that T_2 peaks in Samples S1, S2, C1, and C2 vary from 908.5 to 510.9 ms (i.e., 43% relative decrease), from 825.4 to 749.9 ms (i.e., 9% relative decrease), from 1333.3 to 1100.7 ms (i.e., 17% relative decrease), and from 1615.1 to 1333.3 ms (i.e., 17% relative decrease), respectively. The corresponding variation in estimates of dominant pore radius can be quantified by applying Eq. 9, assuming only spherical pores in the rock. We calculated a relative decrease of 55, 13, 31, and 38% in estimates of pore radius in Samples S1, S2, C1, and C2, respectively.

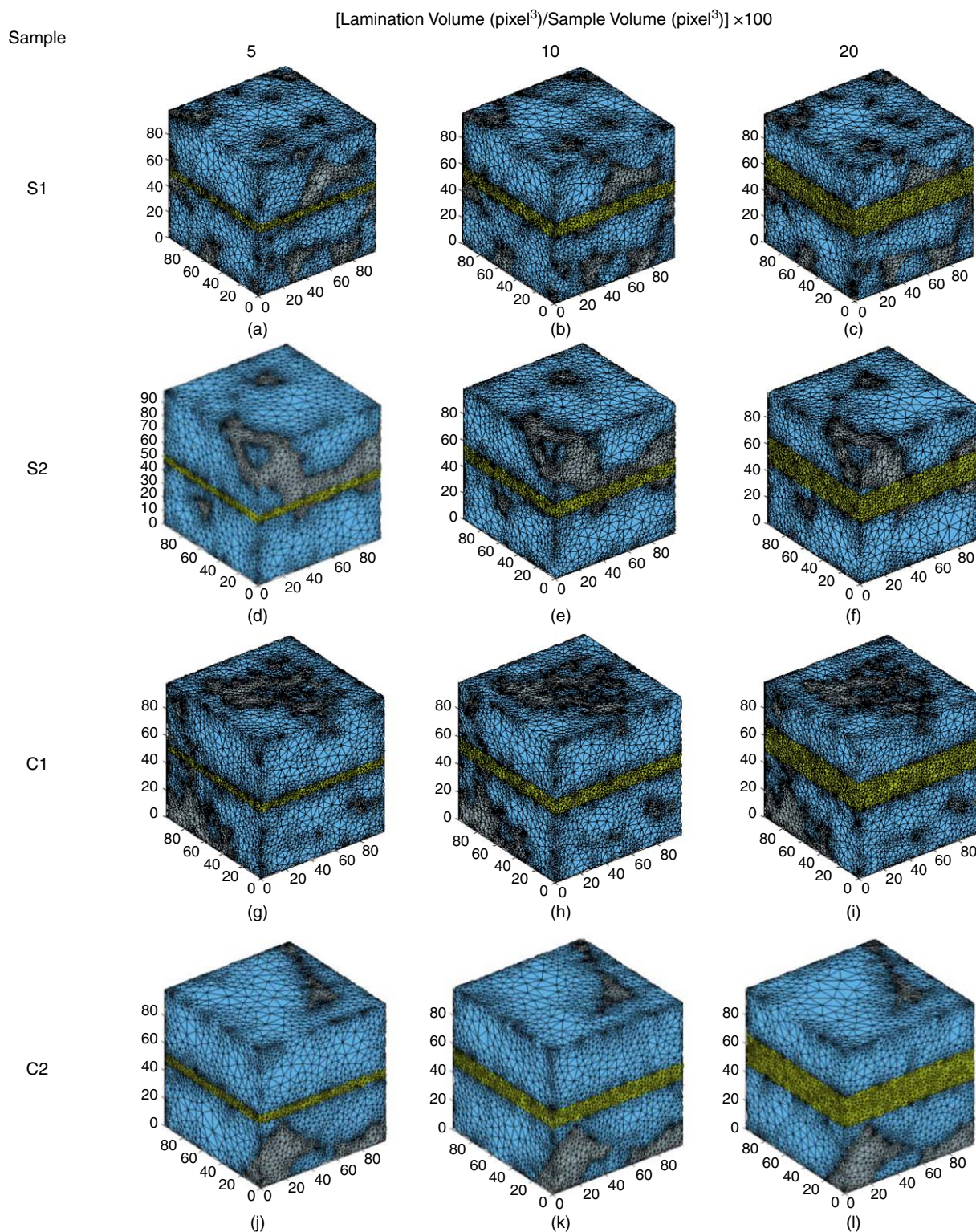


Fig. 2—The 3D voxelized rock Samples S1, S2, C1, and C2, including synthetically introduced laminated-shale layers of different sizes. The blue, gray, and yellow regions represent grains, water-filled pores, and laminated shale, respectively.

Sample No.	Sample Porosity			
	Lamination Volume (pixel ³)/Sample Volume (pixel ³)			
	0%	5%	10%	20%
S1	0.154	0.145	0.141	0.128
S2	0.127	0.111	0.107	0.093
C1	0.179	0.178	0.176	0.159
C2	0.201	0.199	0.195	0.193

Table 3—Rock Samples S1, S2, C1, and C2. Variation in sample porosity with increase in volume of shale lamination synthetically introduced in the rock sample.

Parameter	Value	Parameter	Value
G (T/m)	0.1	$\rho_{\text{sandstone}}$ ($\mu\text{m/s}$)	10
τ (seconds)	0.0005	$\rho_{\text{carbonate}}$ ($\mu\text{m/s}$)	3
D (cm^2/s)	2.5×10^{-5}	ρ_{shale} ($\mu\text{m/s}$)	10
$T_{1B,\text{sandstone}}$ (seconds)	2.5	γ ($\text{MHz} \cdot \text{T}^{-1}$)	42.6
$T_{1B,\text{carbonate}}$ (seconds)	2.5	x_W (vol/vol)	-9.03×10^{-6}
$T_{1B,\text{clay}}$ (seconds)	0.1	x_{SS} (vol/vol)	-16.41×10^{-6}
$T_{2B,\text{sandstone}}$ (seconds)	2.5	x_{cal} (vol/vol)	-13.1×10^{-6}
$T_{2B,\text{carbonate}}$ (seconds)	2.5	x_{illite} (vol/vol)	41.6×10^{-6}
$T_{2B,\text{shale}}$ (seconds)	0.1	B_0 (Gauss)	470

Table 4—Assumed input parameters to numerical simulations of Bloch (1946) equations.

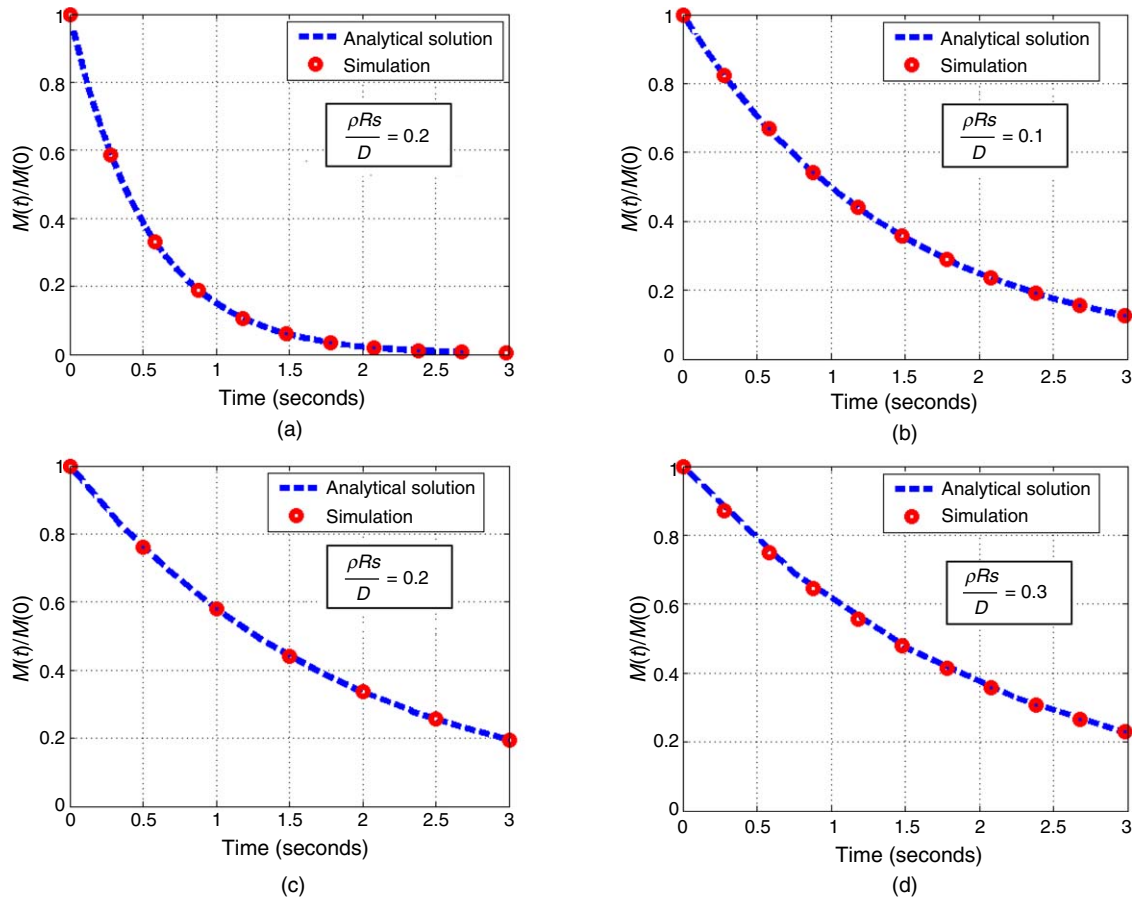


Fig. 3—Comparison of simulated magnetization decays for spherical pores of different pore radii against the analytical solution provided in Brownstein and Tarr (1979) for (a) pore radius = 10 μm ; (b) pore radius = 50 μm ; (c) pore radius = 100 μm ; and pore radius = 150 μm . The blue line shows the analytical magnetization decay, and the red dots show the simulated magnetization decay.

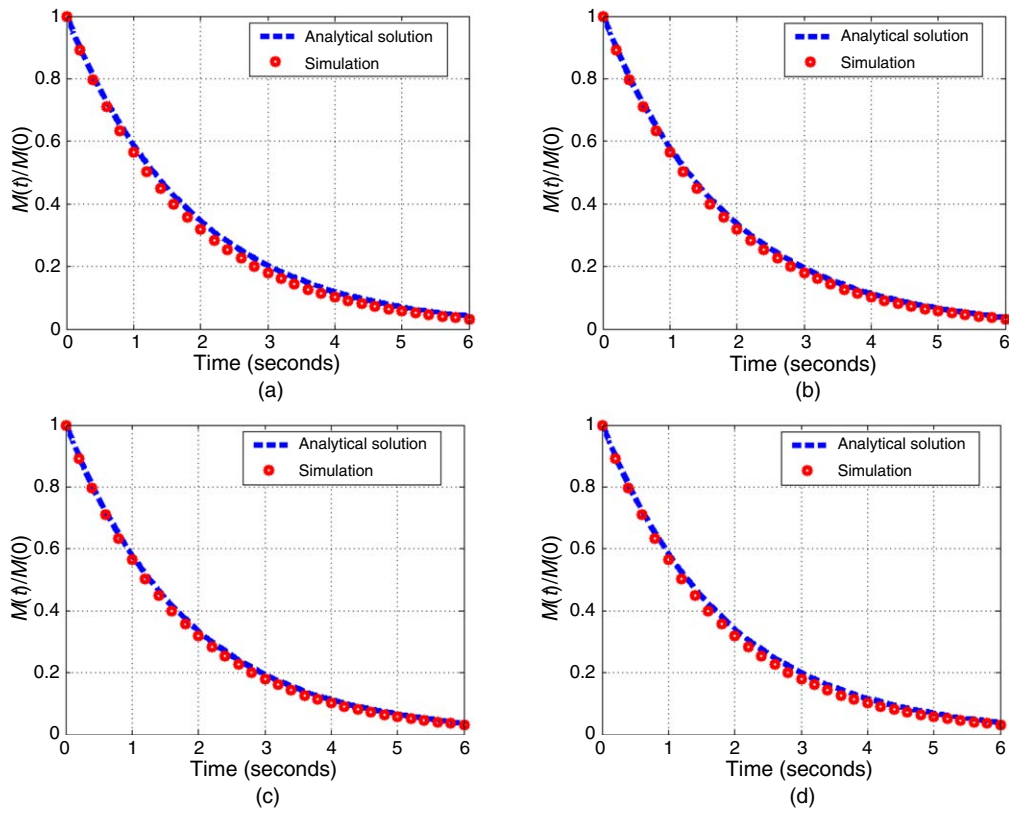


Fig. 4—Comparison of simulated magnetization decays for spherical pores of different pore radii against the analytical solution provided in Kleinberg and Horsfield (1990) for (a) pore radius = 10 μm ; (b) pore radius = 50 μm ; (c) pore radius = 100 μm ; and (d) pore radius = 150 μm . The blue line shows the analytical magnetization decay, and the red dots show the simulated magnetization decay.

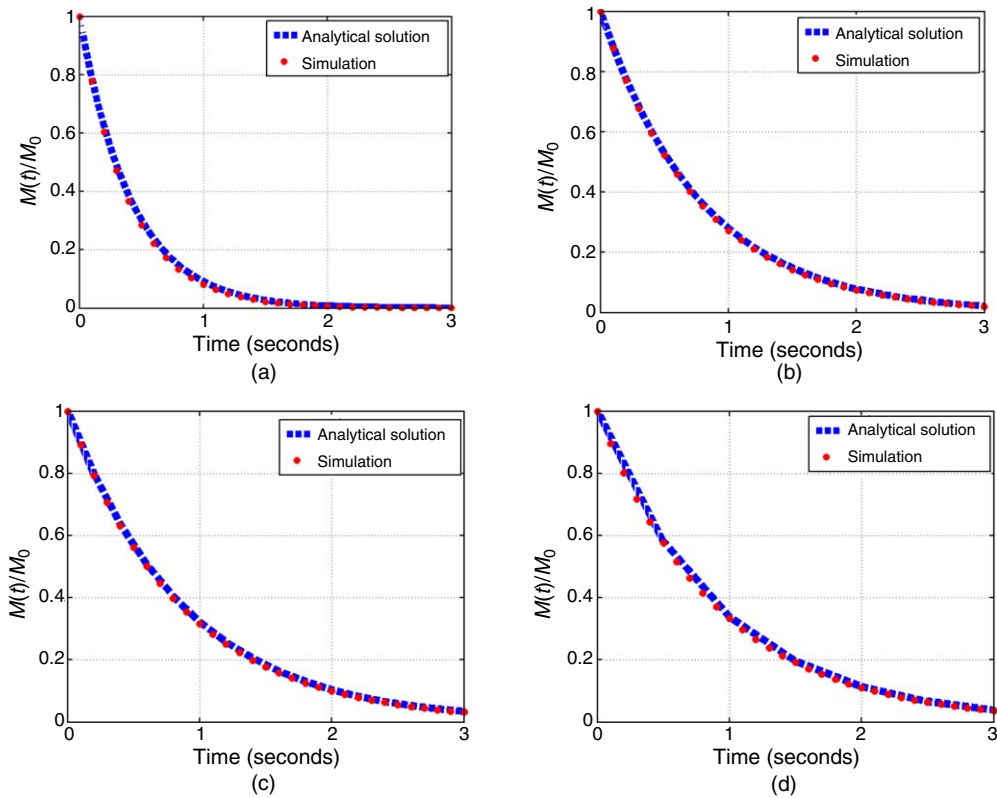


Fig. 5—Comparison of simulated magnetization decays for spherical pores of different pore radii against the analytical solution provided in Brownstein and Tarr (1979) and Kleinberg and Horsfield (1990) for (a) pore radius = 10 μm ; (b) pore radius = 50 μm ; (c) pore radius = 100 μm ; and (d) pore radius = 150 μm . The blue line shows the analytical magnetization decay, and the red dots show the simulated magnetization decay.

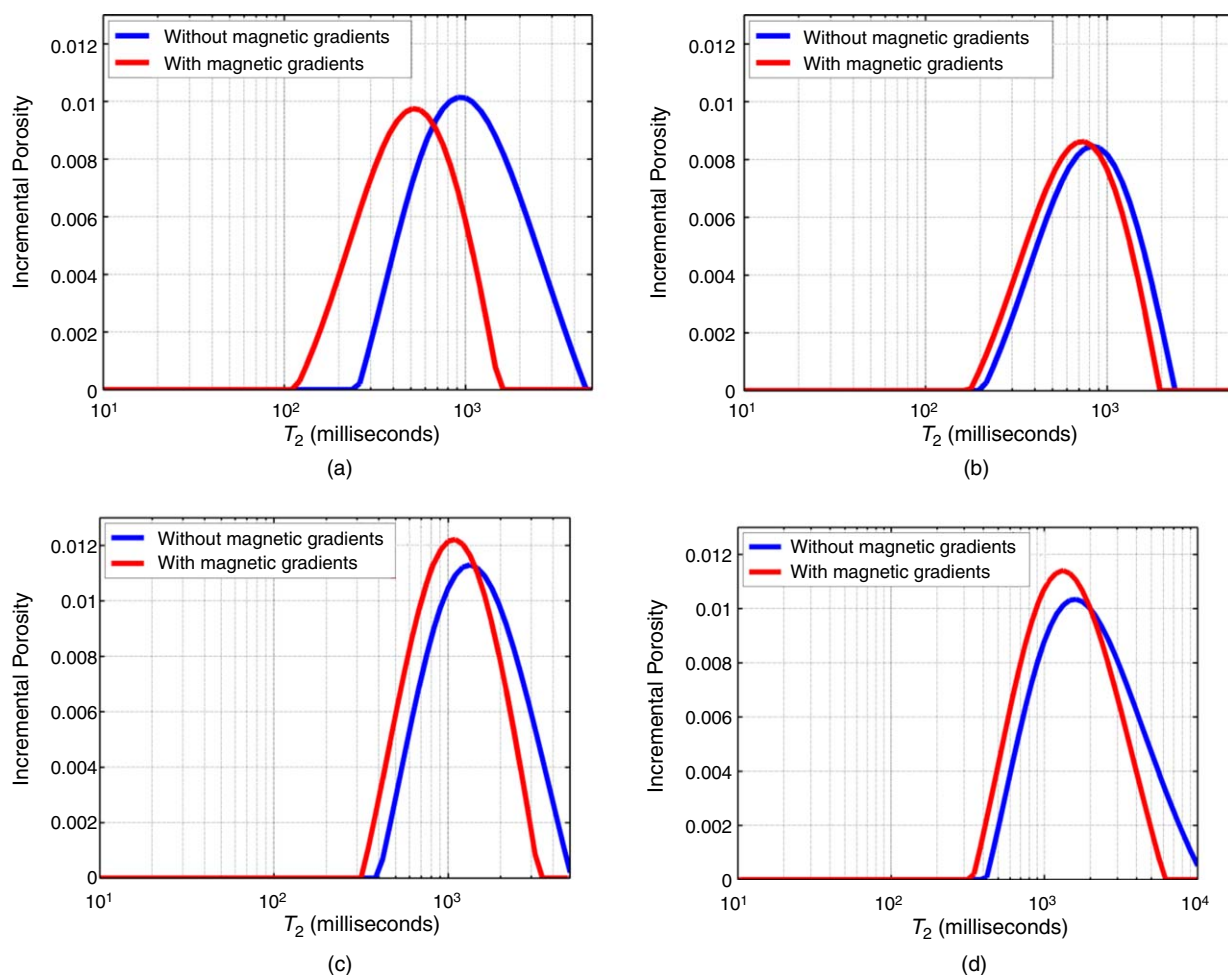


Fig. 6—Simulated T_2 distribution showing the effect of external magnetic-field gradients in rock Samples (a) S1, (b) S2, (c) C1, and (d) C2. The blue and red lines represent the T_2 distributions obtained before and after taking into account the externally applied magnetic-field gradients, respectively.

The effect of internally generated static magnetic gradients depends not only on the pore-size distribution but also on the area of the sample exposed to the fluid, as well as diffusional coupling in the pores. We observe that effect when we calculate the overall volumetric clay minerals in the case of dispersed shale (Table 2). In both cases, the shale layer is assumed to have the same thickness and the same concentration of clay (i.e., 60%). The concentration of overall volumetric concentration of clay calculated for Sample S1 is close to 8% but for Sample S2 is close to 5%. This difference arises because Sample S1 has much more area exposed to the fluid. Hence the size-distributions might look similar, but the relative shifts in T_2 distributions are different for the applied magnetic-field gradient.

Effect of Pore-Fluid/Rock-Grain Susceptibility Contrast on NMR T_2 Distribution. We simulated magnetization decay by considering susceptibility contrast between grains and pore fluids in Sample S1. **Fig. 7** shows the simulated T_2 distribution in this sample. We observed that even at echo time of 1 ms, there is negligible sensitivity of the T_2 peak to the internal magnetic-field gradients. Similar results were achieved for Samples S2, C1, and C2.

Effect of Internal Magnetic-Field Gradients Associated With Dispersed Shale on NMR T_2 Distribution. **Fig. 8** shows simulated T_2 distributions in Samples S1, S2, C1, and C2 demonstrating the effect of internal magnetic-field gradients generated from dispersed shale in each sample on NMR measurements. The blue line shows the T_2 distribution with shale-layer thickness of 0 nm. The T_2 distributions in **Fig. 8** are obtained in the presence of external magnetic gradients and the internal magnetic-field gradient generated from the existing shale.

The presence of a thin layer of shale (e.g., 500 nm) has a negligible effect on T_2 response in all the samples, but as the thickness of the shale layer on grain surfaces increases to 5000 nm, we observe a significant shift in the locations of T_2 peaks in all the samples. The results showed a decrease in T_2 peak from 510.9 to 261.0 ms (i.e., 49% relative decrease), from 749.9 to 421.7 ms (i.e., 43% relative decrease), from 1100.7 to 510.9 ms (i.e., 53% relative decrease), and from 1333.3 to 908.5 ms (i.e., 32% relative decrease) for Samples S1, S2, C1, and C2, respectively. The decrease in T_2 peaks results in a relative decrease of 54, 52, 67, and 50% in the assessment of dominant pore radius in Samples S1, S2, C1, and C2, respectively.

Effect of Internal Magnetic-Field Gradients Associated With Laminated Shale on NMR T_2 Distribution. We finally examined the effect of the presence of shale laminations on NMR T_2 distribution. We synthetically generated the cases with variable volumetric concentrations of laminated shale (Table 3), with clay concentration of 60% in shale laminations. **Fig. 9** shows the simulated T_2 distribution, illustrating the effect of internal gradients generated from clays randomly distributed on shale laminations in Samples S1, S2, C1, and C2. Sample S1 shows a T_2 peak shift from 510.9 to 464 ms (i.e., 9.9% relative change) and Sample S2 shows a T_2 peak shift

from 749.9 to 618.9 ms (i.e., 17.5% relative change) as the lamination volumetric concentration increases from 0 to 20% in sandstone samples. We observe no shift in the T_2 peak location in Sample C1 and a T_2 peak shift from 1333.3 to 908.5 ms (i.e., 31.8% relative change) in Sample C2 as the lamination volumetric concentration increases from 0 to 20% in carbonate samples. The relative decrease in estimates of dominant pore radius is 9.9, 23.1, and 50.0% in Samples S1, S2, and C2, respectively.

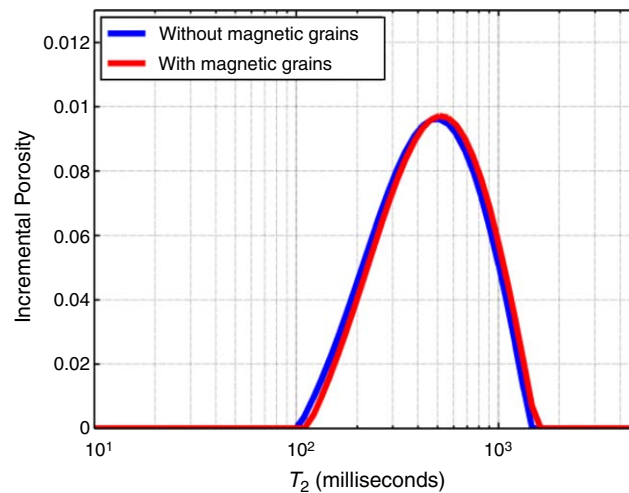


Fig. 7—Simulated T_2 distribution showing the effect of internal magnetic-field gradients generated by magnetic grains in sample S1. The blue and red lines represent the T_2 distributions obtained before and after taking into account the internal magnetic-field gradients generated by magnetic grains, respectively.

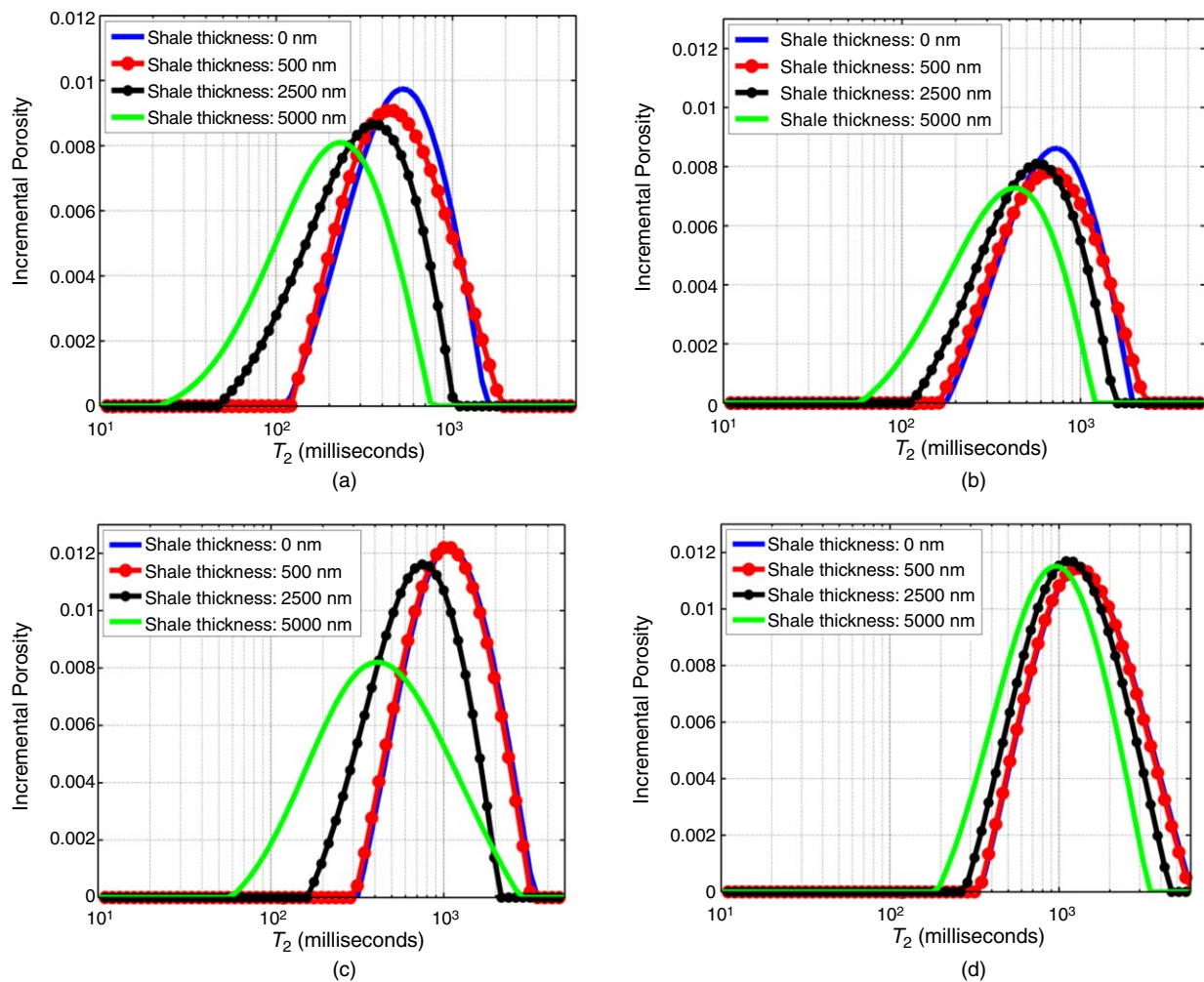


Fig. 8—Simulated T_2 distributions showing the effect of internal magnetic-field gradients generated from dispersed shale in Samples (a) S1, (b) S2, (c) C1, and (d) C2. The blue line, red dots, black line, and green line show the T_2 distributions for shale-layer thicknesses of 0, 500, 2500, and 5000 nm, respectively.

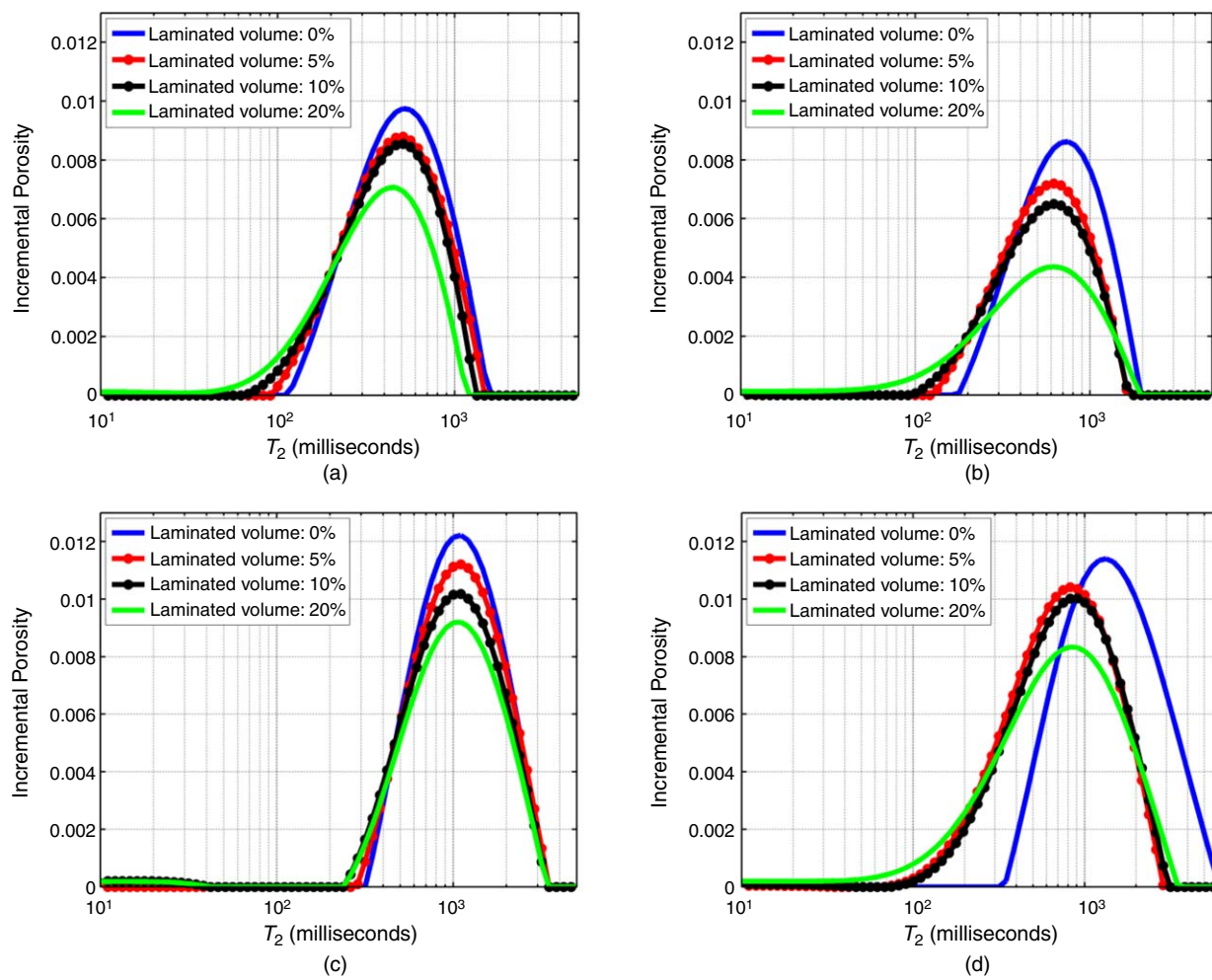


Fig. 9—Simulated T_2 distribution showing the effect of internal magnetic-field gradients generated from shale laminations in Samples (a) S1, (b) S2, (c) C1, and (d) C2. The blue, red, black, and green lines represent shale-lamination volumes of 0, 5, 10, and 20% of the volume of the initial samples, respectively.

Discussions

Table 5 provides a summary of results for dominant peak sizes obtained using simulations for different distributions of magnetic-field gradients discussed in this text. External magnetic-field gradients in a sample cannot be avoided because they are the nature of the logging tool or inside-out spectrometer used for making NMR measurements. The effects of external gradients are more significant because they affect how a proton spins homogeneously in the entire pore space and lead to increased proton dephasing.

Sample	With No Magnetic Gradients	With Only Externally Applied Gradients	Dominant T_2 Peak (milliseconds)					
			Thickness of Shale Layer on Grain Surface			Lamination Volume (pixel ³)/Sample Volume (pixel ³)		
			500 nm	2500 nm	5000 nm	5%	10%	20%
S1	908.5	510.9	464.0	368.7	261.0	464.0	464.0	464.0
S2	825.4	749.9	681.9	562.4	421.7	618.9	618.9	618.9
C1	1,333.3	1,100.7	1100.7	746.9	510.9	1,100.7	1,100.7	1,100.7
C2	1,615.1	1,333.3	1100.7	1000.0	908.5	908.5	908.5	908.5

Table 5—Rock Samples S1, S2, C1, and C2. Summary of results for location of dominant T_2 peak for different cases of externally applied and internally generated magnetic-field gradients discussed in this text.

We observed no change in T_2 peak caused by the presence of diamagnetic centers in quartz and calcite grains. Similar results are also seen in other studies, including Hirasaki et al. (2001), Flaum (2007), and Anand and Hirasaki (2007), which found a limited effect of internal gradients in both carbonates and clay-free Bentheimer sandstone samples during experiments.

We observed this minimal change because quartz and calcite grains have magnetic susceptibilities of -16.41×10^{-6} vol/vol and -13.1×10^{-6} vol/vol, respectively, which are very close to magnetic susceptibility of water (i.e., -9.03×10^{-6} vol/vol) in the pore space. Furthermore, the diamagnetic centers that are in the middle of the grains do not significantly influence internal gradients because the generated internal magnetic-field decay is inversely proportional to the cube of distance away from the diamagnetic centers.

We observed a gradual decrease in dominant T_2 peak as the thickness of the dispersed-shale layer increases. Similar decreases in the dominant T_2 peak were obtained for silica grains in the experimental work of Anand (2007) and Anand and Hirasaki (2007). The effects of internally generated magnetic-field gradients are usually local. A layer of magnetic centers on the surface of grains creates large magnetic-field gradients in their vicinity. The changes in T_2 peaks are gradual because most of the centers are close to the pore fluid and the magnitude of gradient generated depends only on the concentration of magnetic particles.

In addition to internal magnetic-field gradients, the presence of the dispersed shale in the rock causes a shift in T_2 distribution by changing the pore-size distribution and blocking of pores and pore throats; creating variations in surface relaxivity and surface/volume ratios of shale dispersed on the grains; and imposing diffusive coupling of large grain pores with pores in the shale sample. It is possible that the previously discussed mechanisms completely mask the effect of internal magnetic-field gradients generated by paramagnetic clays. This study isolates the effect of only internal magnetic-field gradients on NMR response in the rock samples. We demonstrate that the shift in T_2 distribution caused by magnetic-field gradients can be significant in rocks with high concentrations of paramagnetic clays, such as the case of North Burbank sandstone, which was previously documented by Hirasaki et al. (2001) and Anand (2007).

The decrease in the dominant T_2 peak was more abrupt for the cases of laminated shale. Similar results were obtained by Alhwety et al. (2014) using synthetic rock samples. We observed abrupt changes because the magnetic centers are assumed to be randomly distributed in the shale layer. In the case of laminated shale, the shift in the T_2 distribution and the change in rock porosity depend on the location of the lamination and its proximity to the pores. This mechanism is illustrated in **Figs. 10 and 11**, which show the effect of the presence of shale lamination in Samples C1 and C2, respectively. We can clearly see that shale lamination intersected some of the pores in Samples C1 and C2, decreasing their overall porosity. In Sample C2, the shale lamination is also situated close to other pores in the sample, and hence causes movement of the T_2 peak. On the other hand, the pores in Sample C1 are situated farther away from the shale lamination. Hence, the movement in the peak is minimal when the relative volume of the lamination is 5% but there is movement in the peak when the volume increases to 20%. Consequently, even if the concentration of magnetic centers increases, their influence on the T_2 peak might be limited when they are situated away from the pore fluid.

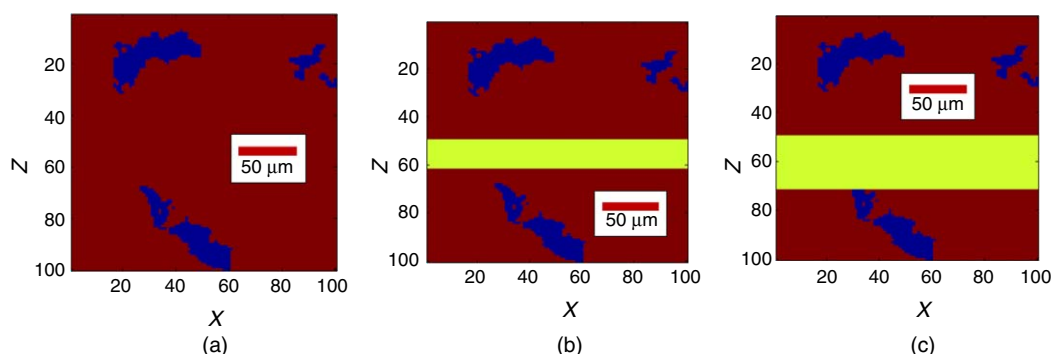


Fig. 10—Sample C1. Presence of shale lamination in (a) the original sample, (b) 10 vol% of laminated shale, and (c) 20 vol% of laminated shale. The red, blue, and yellow regions represent grains, pores, and shale, respectively.

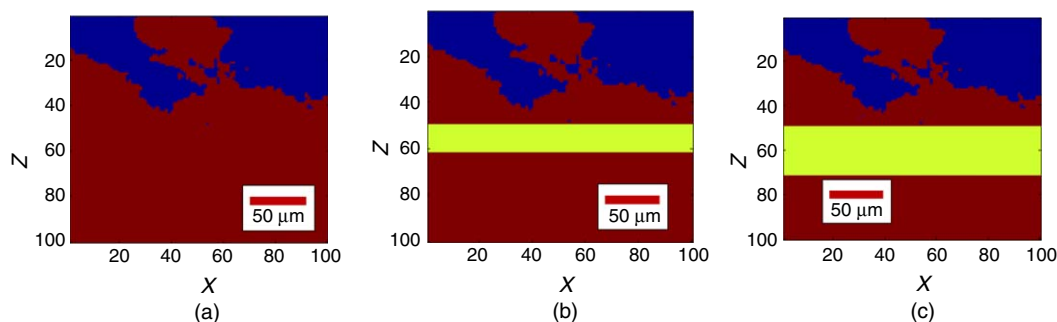


Fig. 11—Sample C2. Presence of shale lamination in (a) the original sample, (b) 10 vol% of laminated shale, and (c) 20 vol% of laminated shale. The red, blue, and yellow regions represent grains, pores, and shale, respectively.

The results of numerical simulations demonstrated the importance of identifying the distribution of magnetic-field gradients during NMR interpretation. Externally applied magnetic-field gradients are usually a characteristic of the spectrometer used for the measurement of NMR response. The effects of these gradients can be mitigated using shorter echo spacing and better tools with smaller externally applied gradients. The effects of internally generated gradients are more difficult to identify and mitigate. As demonstrated in this text, the effects of internally generated gradients depend on shale type, topology, and concentration of magnetic particles. These effects will be accentuated if the shale is dispersed and if it contains high concentrations of paramagnetic clays such as chlorite and illite. Accurate assessment of shale concentration and its magnetic properties is required for reliable interpretation of NMR T_2 measurements. If it is suspected that internal magnetic-field gradients might be affecting the NMR T_2 response, or if the magnetic properties of shale are uncertain, then using tools with smaller echo spacing times or using T_1 response provides a more-reliable method for assessment of rock properties. Use of pore-scale simulation of NMR presents another possible method of obtaining accurate estimates of rock properties in such cases.

The results of numerical simulations depend strongly on echo spacing, initial static magnetic field, and the external magnetic-field gradients used in the NMR numerical simulations. The results shown in this paper are applicable to the specific range of simulation parameters used in this publication. These results might not cover characteristics of all the NMR logging tools and experimental setups. It should also be noted that in our model the pores are assumed to be in a free-diffusion regime. Although this is a valid assumption for most of the pores encountered in sandstones and carbonates, it might cause errors in magnetic-decay calculations in extremely small pores. Shale, clay minerals, and nonclay minerals are also modeled as dipoles using the effective-medium theory, which introduces the limitation of sharp decay in a magnetic field away from the dipole. In addition, the micro-CT imaging process contains the drawback of not resolving pores that are smaller than the resolution of the equipment. Finally, we are sampling small volumes of the rocks for imaging and meshing. In the case of highly heterogeneous rocks, such as carbonates containing vugs and micropores, the pore-scale samples selected for simulation might not represent the configuration of pore space in the formation. Selection of a larger sample size, if computational capabilities allow, can help in capturing pore-scale variations in heterogeneous rocks.

Conclusions

We quantified the effects of externally applied magnetic-field gradients and internally induced static magnetic fields caused by diamagnetic grains or dispersed/laminated-shale distributions on NMR T_2 distribution using pore-scale numerical simulations. To investigate the variation of magnetic-field gradients on different rock samples, we used two sandstone samples and two carbonate samples. The grains and clay particles were modeled as dipoles distributed on the grain surface, inside the rock grains, or within shale laminations. We used a theoretical effective-medium model developed for dipoles in a constant magnetic field to investigate the effects of the grain/pore-fluid and clay/pore-fluid susceptibility contrast on NMR response. We observed significant changes in NMR T_2 distribution in the presence of dispersed and laminated magnetic shale and observed minimal changes caused by diamagnetic quartz and calcite grains.

This paper illustrated the use of pore-scale simulations in developing a fundamental understanding of the behavior of NMR in situations where internal gradients generated by susceptibility contrast cannot be ignored. It is generally accepted that the dominant peak of the NMR T_2 distribution represents the dominant pore size of the rock; however, we demonstrated that the presence of significant internal gradients could lead to an underestimation in the size of dominant pores, pore-size distribution, and rock permeability using NMR-based formulations such as the Timur-Coates model (Coates, G. R., Xiao, L., and Prammer, M. G. 1999).

The outcomes of this paper also help in understanding and quantifying the influence of internal gradients caused by the presence of clay minerals and their spatial distribution on NMR measurements, and in improving real-time NMR-based assessment of porosity and pore-size distribution in the presence of significant internal gradients.

Nomenclature

B	= total magnetic field vector, Tesla
B_0	= initial static magnetic field, Tesla
B_{eff}	= external applied magnetic field, Tesla
D	= diffusion tensor, m^2/s
D	= diffusion constant, m^2/s
f	= Larmour frequency, MHz
G	= external magnetic-field gradient, T/m
I_n	= intensity function
l	= length scale where the boundary condition operates, m
M	= magnetization, Tesla
M_x, M_y, M_z	= x-, y-, z-component of magnetization, Tesla
M_0	= initial magnetization, Tesla
n	= normal vector to a surface, m
r	= distance of the random point away from the sphere center, m
R_s	= radius of a spherical pore, m
R_0	= radius of a magnetic sphere, m
S	= surface area of the medium, m^2
t	= time, seconds
T_B	= bulk-relaxation-rate tensor, seconds
T_{B1}	= longitudinal bulk relaxation rate, seconds
T_c, T_n	= time constants, seconds
T_D	= diffusion-induced-decay constant, seconds
T_1	= longitude relaxation rate, seconds
T_{1s}	= relaxation rate for the boundary, seconds
T_2	= transverse relaxation rate, seconds
T_{2B}	= transverse bulk-relaxation rate, seconds
V	= volume of the medium, m^3
x_C	= magnetic susceptibility of calcite, vol/vol
x_{illite}	= magnetic susceptibility of illite, vol/vol
x_{SS}	= magnetic susceptibility of quartz, vol/vol
x_w	= magnetic susceptibility of water, vol/vol
Z	= distance parallel to z-coordinate of the random point, m
γ	= gyromagnetic ratio, MHz/T
μ_g	= magnetic permittivity of the magnetic sphere
μ_w	= magnetic permittivity of the outside medium
ρ	= surface relaxivity, $\mu\text{m}/\text{s}$
$\rho_{\text{carbonate}}$	= surface relaxivity of calcite, $\mu\text{m}/\text{s}$
$\rho_{\text{sandstone}}$	= surface relaxivity of quartz, $\mu\text{m}/\text{s}$
ρ_{shale}	= surface relaxivity of shale, $\mu\text{m}/\text{s}$
τ	= half-echo time, seconds

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Appendix A—Analytical Solution for Surface Relaxation in a Spherical Pores

Brownstein and Tarr (1979) provided the analytical solution to magnetization decay in a spherical pore of radius R_s and surface relaxation ρ , filled with a fluid of diffusion constant D , and transverse bulk relaxation T_{2B} , in the absence of any internal or external field gradients. The analytical solution to the Bloch (1946) equations in this simplified case is given by

$$M = M_0 \sum_1^n I_n \exp\left(-\frac{t}{T_n}\right), \quad \dots \quad (\text{A-1})$$

$$\frac{1}{T_n} = \frac{D\varepsilon_n^2}{R_s^2} + \frac{1}{T_{2B}}, \quad \dots \quad (\text{A-2})$$

and

$$I_n = \frac{12 (\sin\varepsilon_n - \varepsilon_n \cos\varepsilon_n)^2}{\varepsilon_n^2 [2\varepsilon_n - \sin(2\varepsilon_n)]}, \quad \dots \quad (\text{A-3})$$

where T_n is time constant, I_n is the intensity function for each time constant T_n , and ε_n are the positive roots of

$$1 - \varepsilon_n \cot\varepsilon_n = \frac{\rho R_s}{D}. \quad \dots \quad (\text{A-4})$$

Appendix B—Analytical Solution for CPMG Sequence in Spherical Pores

Neuman (1974) and Kleinberg and Horsfield (1990) provide the analytical solution for the spin-echo dephasing in a restricted geometry. For a spherical pore of radius R_s , containing a fluid with gyromagnetic ratio γ , diffusion constant D , and transverse bulk relaxation T_{2B} , under a constant external magnetic-field gradient G and echo time 2τ , the analytical solution is given by

$$M = M_0 \exp\left(-\frac{t}{T_c}\right), \quad \dots \quad (\text{B-1})$$

$$\frac{1}{T_c} = \frac{1}{T_{2B}} + \frac{2\gamma^2 G^2 R_s^4}{D} \sum_{n=1}^{\infty} \left[\frac{1}{\varepsilon_n^4 (\varepsilon_n^2 - 2)} \right] \left[1 - \frac{3 - 4\exp(-b/2) + \exp(-b)}{b} \right], \quad \dots \quad (\text{B-2})$$

and

$$b = \frac{2\tau\varepsilon_n D}{R_s^2}, \quad \dots \quad (\text{B-3})$$

where T_c is the time constant and ε_n are the roots of

$$\tan\varepsilon_n = \frac{2\varepsilon_n}{2 - \varepsilon_n^2}. \quad \dots \quad (\text{B-4})$$

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